



Lab 01: Basic LAN Setup

2911 Router · 2x 2960 Switches · PC0 & PC1

Lab #01

Topic: Networking
Fundamentals

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Build a LAN using two Cisco 2960-24TT switches, a 2911 router and two PCs.
- Assign IP addresses and subnet masks to both router interfaces (Gig0/0 and Gig0/1).
- Configure static IP addresses, subnet masks and default gateways on PC0 and PC1.
- Bring up all router interfaces using **no shutdown** and verify Layer 3 connectivity.
- Relate to Network+ objective 2.2: Compare and contrast networking devices.

2. Network Topology

Router0 (2911) connects to Switch0 (2960-24TT) via GigabitEthernet0/0 and to Switch1 (2960-24TT) via GigabitEthernet0/1. PC0 plugs into Switch0 and PC1 into Switch1. Each router interface is the default gateway for its subnet.

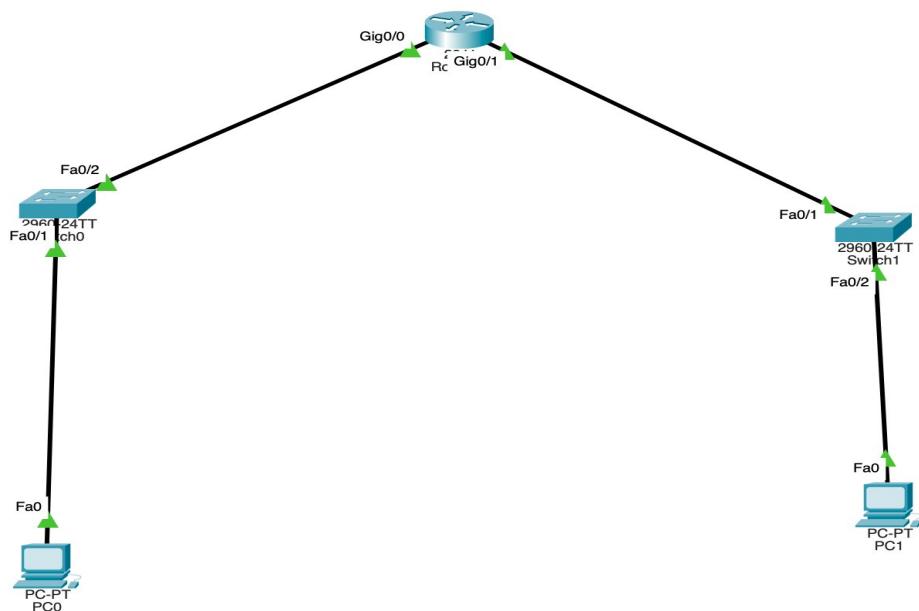


Figure 1 – Router0 (2911) with Gig0/0 → Switch0 → PC0 and Gig0/1 → Switch1 → PC1

3. IP Addressing Table

Every device that needs an IP address is listed below with its full address, subnet mask and default gateway:



Device	Interface	IP Address	Subnet Mask	Default Gateway	Role
Router0	GigabitEthernet0/0	192.168.1.1	255.255.255.0	—	LAN A gateway
Router0	GigabitEthernet0/1	192.168.2.1	255.255.255.0	—	LAN B gateway
Switch0	—	—	—	—	Layer 2 only
Switch1	—	—	—	—	Layer 2 only
PC0	Fa0	192.168.1.10	255.255.255.0	192.168.1.1	Client LAN A
PC1	Fa0	192.168.2.10	255.255.255.0	192.168.2.1	Client LAN B

3.1 Subnet Summary

Subnet	Network Address	Subnet Mask	Gateway (Router)	PC Address	Broadcast
LAN A	192.168.1.0	255.255.255.0	192.168.1.1	192.168.1.10	192.168.1.255
LAN B	192.168.2.0	255.255.255.0	192.168.2.1	192.168.2.10	192.168.2.255

4. Step-by-Step Configuration

4.1 Router0 – Assign IPs to Both Interfaces

Access Router0 via CLI. Enter global configuration mode and assign an IP address and subnet mask to each GigabitEthernet interface, then bring each up:

```
Router>enable
Router#configure terminal
Router(config)#hostname Router0

! --- Interface Gig0/0 facing Switch0 / LAN A ---
Router0(config)#interface GigabitEthernet0/0
Router0(config-if)#ip address 192.168.1.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit

! --- Interface Gig0/1 facing Switch1 / LAN B ---
Router0(config)#interface GigabitEthernet0/1
Router0(config-if)#ip address 192.168.2.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit

Router0(config)#exit
Router0#write memory
```

4.2 PC0 – Static IP Configuration

On PC0: Desktop tab → IP Configuration → Static. Enter the following:



IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	(leave blank for this lab)

4.3 PC1 – Static IP Configuration

On PC1: Desktop tab → IP Configuration → Static. Enter the following:

IP Address	192.168.2.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1
DNS Server	(leave blank for this lab)

5. Verification

Run the following commands to confirm full connectivity and correct configuration:

! On PC0 Command Prompt:

```
PC0>ping 192.168.1.1      ! PC0 to its own gateway (Router Gig0/0)
```

```
PC0>ping 192.168.2.1      ! PC0 to Router Gig0/1
```

```
PC0>ping 192.168.2.10     ! PC0 to PC1 across subnets
```

! On Router0 CLI:

```
Router0#show ip interface brief
```

```
! Gig0/0 must show: 192.168.1.1    up    up
```

```
! Gig0/1 must show: 192.168.2.1    up    up
```

```
Router0#show ip route
```

```
! C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
```

```
! C 192.168.2.0/24 is directly connected, GigabitEthernet0/1
```

```
Router0#show running-config
```

```
! Verify ip address lines under both interfaces
```

6. Key Takeaways

- Every router interface needs its own IP address and subnet mask — this is what makes it a gateway.
- The subnet mask 255.255.255.0 (/24) gives 254 usable host addresses per subnet.
- PC default gateway must match the router interface IP on the same subnet exactly.
- Cisco router interfaces are **shutdown by default** — always use **no shutdown** to activate.
- **show ip interface brief** is the fastest way to verify IPs and interface states (up/up).
- Connected routes are added to the routing table automatically when an interface is configured and up.



- Two separate /24 subnets = two separate broadcast domains — the router bridges them at Layer 3.